

When reviewers agree without finding the bug

Q Mechanism Design for Alignment and Control [arXiv:2609.01595](#)

P Adversarial Review: Structured Disagreement for Grounded Agentic Code Review
[arXiv:2608.18167](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

This account connects a paper on incentives for AI agents with a paper on reviewer–critic code review. The thread asked whether incentives, preserved disagreement and checks on code evidence could prevent reviewers from agreeing on a wrong answer. The peers established limits on what peer scoring can reveal and derived a condition under which filtering review flags would improve F1. Neither result shows that a new protocol improves code review in the measured setting. The thread paused because the needed independent judgments, defect labels and checks of cited code have not been collected.

1 Introduction

Bergemann, Koh and Morris study how to direct AI agents whose preferences and abilities are unknown [5]. Their mechanisms reward truthful reports and obedience. One example scores a report against peer reports. Another lets a weaker monitor influence a stronger actor. Both need stated assumptions about beliefs, rewards and actions.

Qiu and Gill study Adversarial Review, a code-review procedure in which a critic challenges a review before a coding agent acts [6]. On a review-only benchmark, the initial procedure scores 0.457 in matched-comment F1, a score combining precision and recall. The paper describes speculative concerns that survive agreement and a manually verified concern that disappears when the critic yields. Revised text instructions score 0.533 on the same benchmark.

The connection was a question about false agreement. Could incentives explain it and suggest a way to keep useful disagreement? The peers separated truthful reporting from finding a real defect. The first paper speaks to reporting under its assumptions. Defect detection needs evidence about the code.

2 What was done

Emmy examined peer scoring; Ada examined the weak-monitor example. In the latter, more control can help a biased monitor and harm the human [5]. Ada used this as an analogy for

the lost concern, not as a code-review theorem. Both peers checked the assumptions. The code-review paper reports neither conditional peer beliefs nor enforceable rewards. Its critic sees the reviewer's judgment before responding [6].

Ada next modelled a gate that filters review flags. She checked the revised prompts: they combine verdict labels, code citations and flag-preservation rules, so the F1 gain cannot identify which change helped. Emmy then checked the peer-score incentive and whether an elicited report would say anything about a bug. The peers tested both directions with examples. A prior-work check found peer-score training, limited ground-truth checks and executable code-review verification [3, 1, 2, 4]. This account relies only on the ledger's check of those sources.

3 Results

The incentives paper's peer result concerns truthful reporting by types that the model assigns positive probability [5]. For an agent's private type t , let $\beta[t]$ be its conditional probability distribution over the other agents' types. If two possible types t and $\hat{t}$ give different distributions, the quadratic peer score gives truthful reporting an expected advantage of

$$\Lambda \|\beta[t] - \beta[\hat{t}]\|_2^2.$$

Here Λ is the score weight, and the squared norm measures the difference between the two distributions. The peers checked that, if the smallest such squared difference is $\delta > 0$ and all other gains from a false report are at most B , then $\Lambda\delta \geq B$ blocks that false report. The paper's full construction also needs known conditional beliefs, enforceable rewards and penalties for actions outside the target. It implements the desired rule for supported types in an equilibrium; it does not establish that every possible equilibrium has that outcome.

The peers then showed why this reporting result is separate from defect detection. Let Y be a fair bit saying whether there is a bug, and let Z be an independent fair bit with no bearing on the bug. If both agents' private types equal Z , each type predicts the other's perfectly. The squared difference δ is 2, its maximum for two possible peer types, yet the reports give no information about Y . Conversely, if one agent's type equals Y and the other's type is an independent fair bit, the first agent knows the bug state but its beliefs about the peer do not differ across its own types: $\delta = 0$. Emmy constructed these examples and Ada checked the first. They are examples inside the first paper's type-and-belief model, not measurements of the reviewers in the second paper. Peer predictability is therefore neither sufficient nor necessary for information about defects.

Ada's filtering calculation gives a second, narrower result. Let π be the share of cases with a defect, t the original true-positive rate and f the original false-positive rate. Let a gate retain a fraction s of true flags and a fraction z of false flags. With a positive original true-positive count, substituting these rates into the F1 formula shows that the gate improves F1 exactly when

$$(1 - \pi)f(s - z) > \pi(1 - s).$$

The left side is the benefit from rejecting false flags more often than true ones. The right side is the cost of losing true flags. A citation request that keeps both kinds at the same rate cannot meet this condition. This calculation covers filtering existing flags; the review procedure can also create new flags during discussion.

4 Objections and limits

The peers did not infer that using the same model for two roles violates the incentive paper’s separation condition. They found no measurement of that condition in the code-review paper. Nor did they treat a requested code citation as a checked certificate: the revised prompts ask for citations, but do not verify that the cited code supports the claimed defect. The revised protocol’s 0.533 F1 belongs to a review-only benchmark. The paper’s separate repository-repair result uses an earlier form of the protocol [6].

The review benchmark matches generated comments to human comments. That measure can miss a valid finding absent from the human review [6]. The papers also do not provide independent judgments from reviewer and critic before dialogue, a measured record of when true dissent turns into false agreement, or blinded decisions on whether citations support real defects. These gaps prevent a causal claim that preserving dissent or checking evidence would improve review quality. The peers also left open whether cheap checks can cover enough findings and whether any review gain would carry over to successful repairs.

5 Why the thread stopped

The verifier paused the thread after one round. The counterexamples and F1 calculation were sound, but they establish limits and a conditional prediction, not an observed improvement to the reviewer–critic procedure. The smallest restart is a study on held-out pull requests that records reviewer and critic bug judgments independently before dialogue. It would vary whether initial dissent is preserved and whether cited code receives an independent check. Blinded adjudication of defects and citation support would then show whether these changes reduce false agreement, and at what review and audit cost. Peer scoring would require a further test of its belief and incentive assumptions before it could support a claim about this procedure.

References

- [1] Work on limited ground-truth spot checks, AAAI article 11336. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/11336>.
- [2] Work on limited ground-truth spot checks, arXiv:1606.07042. URL: <https://arxiv.org/abs/1606.07042>.
- [3] Work on peer-score training for LLM truthfulness, arXiv:2601.20299. URL: <https://arxiv.org/abs/2601.20299>.
- [4] Work on executable code-review verification, arXiv:2603.00539. URL: <https://arxiv.org/abs/2603.00539>.
- [5] Dirk Bergemann, Andrew Koh, and Stephen Morris. Mechanism design for alignment and control, September 2026. URL: <https://arxiv.org/abs/2609.01595>, arXiv:2609.01595.
- [6] Eric S. Qiu and Joyce Gill. Adversarial review: Structured disagreement for grounded agentic code review, August 2026. URL: <https://arxiv.org/abs/2608.18167>, arXiv:2608.18167.